

Influence of Water on High-Temperature CO₂ Capture Using Layered Double Hydroxide Derivatives

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Mg–Al–CO₃ layered double hydroxide (LDH) was synthesized, and its thermal evolution was investigated using X-ray diffraction FTIR techniques. These patterns revealed the phase transformation of crystalline LDH to a nearly amorphous layered double oxide (LDO). LDOs derived after calcination at 400 °C showed good sorption potential for CO₂, especially in the context of high-temperature CO₂ separation from flue gases. Presence of water in the feed proved to have a positive effect as CO₂ sorption increased from 2.72% (0.61 mmol/g) to 3.14% (0.71 mmol/g) tested at dry- and wet-gas conditions, respectively. CO₂ sorption studies conducted using wet mixed gas (14% CO₂) have also shown high sorption capacity even though CO₂ concentration was diluted by almost seven times. Temperature cycling in wet conditions demonstrated high levels (75%) of desorption, which reached an equilibrium value (67%) after initial stabilization. Shorter time cycles (10 min) were found to be more effective in improving the overall efficiency of the process. Regeneration of the LDOs at 400 °C retrieved more than 90% of the original sorption capacity.

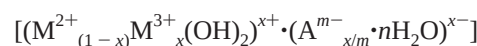
Introduction

Combustion of fossil fuels such as coal, natural, gas and petroleum products for power generation and transportation have increased CO₂ emissions drastically in the recent past. In this context, the separation and capture of CO₂ from gas streams is becoming increasingly significant. There are several existing technologies available for CO₂ capture such as solvent absorption, adsorption, membrane, and cryogenic separation.^{1–4} However, each of these methods has their own limitations based on operating temperatures and pressures, and materials, though absorption by amine solvents, have been extensively used by the gas and petrochemical industries. Because the temperature of flue gases are in excess of 100 °C, efficient CO₂ capture by conventional technologies require gas cooling, which translates in further energy penalties in power generation.

Keeping these issues in mind, materials research became paramount to develop suitable inorganic compounds, which possess high selectivity and good sorption capacity for CO₂ at relatively high temperature. In addition, adequate sorption/desorption kinetics, stable sorption capacity, and good mechanical strength of the sorbent solvents/particles are desirable.^{5,6} Inorganic materials such as activated carbons and zeolites (13X, LSX, and type A) can address some of these requirements, though for low-temperature (<100 °C) CO₂ separation application.^{7,8} However, zeolites are only effective for CO₂ separation from the gas mixtures containing species that are less polar than CO₂. Hence, CO₂ adsorption capacities of zeolites were found to be severely influenced by the presence of water and SO₂.^{5,8–11} Activated carbons were also widely investigated as carbon dioxide adsorbents because of their high surface area, which confers them high adsorption capacity.^{9–11} However, both zeolites and activated carbons were found not attractive for high temperature applications because the CO₂ adsorption capacity of these materials drastically decreases with temperature. For example, ASRT-5A zeolites showed a very high CO₂ adsorption

potential equivalent to 15.8% by weight at 25 °C, which subsequently decreased to 1.43% at 250 °C.⁵

To address some of the shortcomings of conventional processes, a new class of inorganic materials, namely layered double hydroxides (LDHs), has attracted the interest of the scientific community for CO₂-capture applications. Of particular attention, their layered double oxide (LDO) derivatives produced on calcination were found to be hydrothermally stable, possessing desired properties as CO₂ sorbents in post-combustion capture applications. LDHs belong to a large group of anionic or basic clays. LDHs exist as minerals with names depending on their metal-ion composition. For example, Mg–Al LDHs are referred as hydrotalcite-like compounds (HTLCs). Their structure consists of positively charged brucite-like (Mg(OH)₂) layers with trivalent cations substituting for divalent cations at the centers of octahedral sites of hydroxide sheet. The excess positive charge in the main layers of LDHs is compensated by anions and water molecules in the interlayer regions. LDHs are represented by a general formula,



where M²⁺ = Mg²⁺, Ni²⁺, Zn²⁺, Cu²⁺, Mn²⁺, and so forth, and M³⁺ = Al³⁺, Fe³⁺, Cr³⁺, and so forth, and A^{m-} = CO₃²⁻, SO₄²⁻, NO₃⁻, Cl⁻, OH⁻. The value of x is normally between 0.17 and 0.33, but there is no strict limitation of this ratio.¹²

Heat treatment of crystalline LDHs at higher temperatures leads to important structure evolutions and phase changes. According to Constantino et al.¹³ crystalline LDH changes to amorphous LDO phase upon heating to 400–600 °C. The amorphous LDO phase of LDH was identified as most appropriate for CO₂ capture. LDOs have potential applications as CO₂ sorption materials to take advantage of acid–base interactions between acidic CO₂ and basic sites on LDOs. Several research groups^{14–20} have investigated CO₂ adsorption on LDH and LDO materials. Yong et al.^{17,18} have studied CO₂ adsorption of LDHs at higher temperatures and observed an adsorption capacity of 0.5 mmol/g at 300 °C and 1 bar. They have found decomposition of LDH beyond 300 °C resulting in a micro-

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porous surface with increased surface area and pore volume. This directly enhances the capacity of CO₂ adsorption. Water or steam content was found to favor the adsorption capacity particularly at lower pressures (~ 0.05 bar).¹⁸ Ding et al.¹⁶ have also studied CO₂ adsorption at higher temperatures under steam reforming of methane conditions. At 480 °C and in the presence of water, they found an adsorption capacity of 0.58 mmol/g, which was independent of water vapor content in the feed. A steady decline in the reversible adsorption of CO₂ was also noticed in their experiments.

Flue gases emitted from power stations contain considerable amounts of water in the form of steam. The percentage of water found in the flue gas emitted from different sources varies between 7 and 22% with the emissions from brown coal combustion having maximum water, which may adversely influence CO₂ capture when using conventional solid sorbents (i.e., zeolites). Hence, the study of the sorption of water by LDOs and its influence on CO₂ sorption is very important in the context of CO₂ capture from flue gas, though just a few preliminary studies have been reported^{16–18} so far. The main objective of this work is to investigate how LDOs sorption performance can be influenced under CO₂ wet-gas streams. A detailed study of the sorption of CO₂ by LDO was conducted using both pure CO₂ and a gas mixture with flue-gas composition. CO₂ and water-sorption capacities of LDO were quantitatively differentiated from the combined sorption, and the reversibility of the sorption under both wet and dry conditions was examined for its application in repeated sorption/desorption cycles.

Experimental Section

LDH Preparation. Mg–Al–CO₃ LDH was prepared by the coprecipitation method. 1.0 M solutions of Mg(NO₃)₂·6H₂O and Al(NO₃)₃·9H₂O were mixed together in the required ratio and 200 mL of this solution was added to the base solution containing NaOH and Na₂CO₃. The resultant slurry was aged for 24 h at 80 °C with continuous stirring. LDHs were separated by centrifuging, and the resultant cake was washed with deionized water 5–6 times to completely remove sodium salts from the LDH. LDH was finally dried in an oven at 100 °C overnight.

Structural Characteristics. FTIR, X-ray diffraction (XRD), and thermogravimetric analysis (TGA) techniques were used to identify LDH structure and thermal evolution pattern. A PerkinElmer FTIR Spectrometer (Spectrum 2000) was used to record the infrared spectra, whereas the XRD patterns were recorded using a Rigaku Miniflex X-ray diffractometer (with a variable slit width). A Shimadzu TG50 TGA instrument was used to investigate thermal decomposition patterns of the LDH.

CO₂ Sorption. Thermogravimetric sorption of CO₂ on LDO was measured using a Shimadzu TG50 Thermogravimetric analysis (TGA) instrument. LDH samples were calcined in situ at 400 °C for 4 h in an inert helium atmosphere flowing at 80 mL/min before measuring CO₂ sorption on the resulting LDOs at 200 °C. A constant flow of CO₂ at 80 mL/min was maintained during the CO₂ sorption experiments. To investigate the effect of water on adsorption, CO₂ and helium wet-gas streams were made up by passing each gas through a water bubbler at 50 °C, thus resulting in a water saturation of 12%.¹¹ A synthetic flue gas (mixed gas) with 14% CO₂, 4% O₂, and 82% N₂ was also used in this study to mimic flue-gas composition and measure sorption capacity. CO₂ sorption/desorption behavior of LDOs was investigated using a programmed temperature cycling with single gas as well as mixed-gas systems in the presence of water.

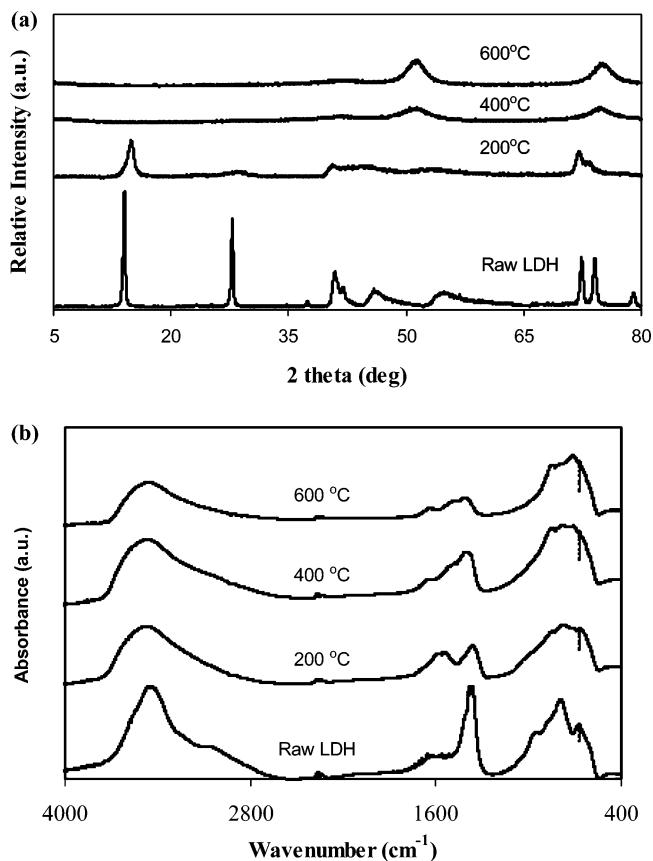


Figure 1. (a) XRD patterns and (b) FTIR of raw LDH and its structure evolution as function of temperature.

In a typical experiment, CO₂ sorption was conducted at 200 °C for 30 min under a feed-gas flow of 80 mL/min followed by desorption at 300 °C for 30 min under helium atmosphere.

Results and Discussion

Structural Aspects of LDH. Thermal evolution of LDH structure into the LDO phase is the critical process because the resultant LDOs are used for CO₂ sorption in this study. XRD and FTIR traces shown in Figure 1 illustrate the formation of crystalline LDH and its transformation to a nearly amorphous LDO phase upon heating the sample to 600 °C. In the XRD patterns of raw LDH (part a of Figure 1), the basal reflections from the (003) and (006) planes and the reflection from plane (110) are indicative of Mg–Al–CO₃ LDH formation with an interlayer spacing of 0.760 nm and a unit cell parameter *a* of 0.305 nm. The presence of carbonate in the interlayer is evidenced by the characteristic vibration at 1360 cm^{−1} in the FTIR spectrum (part b of Figure 1). Further, the broad band at 3400 cm^{−1} with a shoulder at 3100 cm^{−1} and another broad band at 780 cm^{−1} are assigned to the O–H stretching and M–O bond stretching vibrations in the raw LDH phase, respectively. Upon heat treatment, the crystalline LDH phase progressively changes to nearly amorphous LDO with the calcination temperature. The XRD patterns in part a of Figure 1 indicate that the LDH layered structure is partially collapsed after calcination at 200 °C and then completely disappears at 400 °C. The resultant LDOs derived at 400 and 600 °C have two broad peaks at 52 and 75° in XRD patterns (part a of Figure 1), which corresponds to periclase MgO phase (JCPDS 43–1022) being nearly amorphous. The de-CO₂ process continues to calcination temperatures of over 600 °C as the weak IR peak at 1360 cm^{−1} is still observable (part b of Figure 1). The thermal decomposi-

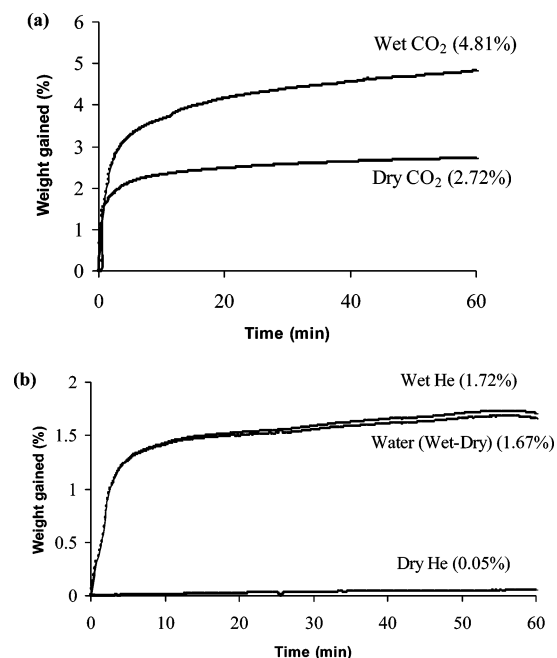


Figure 2. (a) Comparison of sorption of dry and wet CO₂ and (b) sorption of wet helium, dry helium, and water (wet He–dry He) by the LDO.

tion patterns observed in TGA experiments also suggest a series of physical/chemical changes occurring during the calcination process. These reactions are clearly identified as dehydration, dehydroxylation, and decarbonation. The TGA results indicated a weight loss of 14.4% observed below 280 °C, which is mainly attributed to the dehydration process in addition to small amounts of dehydroxylation. The major weight loss 23.8% observed between 280 and 600 °C accounts for the dehydroxylation and decarbonation of LDH. The percentage weight loss and the species lost in different temperature ranges observed in this study are similar to those reported in earlier studies.^{17,20}

Sorption Using Single-Gas (Pure CO₂) Feed. Part a of Figure 2 shows the LDO's dry and wet CO₂ sorption curves measured at 200 °C. Dry CO₂ sorption has a maximum weight gain of 2.72% (0.61 mmol/g) after 60 min, whereas the wet CO₂ sorption increases the weight of the LDO to 4.81%, clearly showing an additional 2.09% sorption gain. To deconvolute the water and CO₂ sorption on LDO, similar experiments at 200 °C were conducted using dry and wet helium, which is a gas with almost no sorption capacity for non-cryogenic temperatures. Part b of Figure 2 shows that helium has almost no significant sorption affinity to LDO, whereas the water-sorption profile of LDO clearly indicates a water weight gain of 1.67% (i.e., sorption difference between wet and dry helium gas streams). This shows that LDOs have tendency to take up a good amount of water almost instantaneously, similar to CO₂ sorption. Therefore, under wet-gas stream conditions, it can be deduced that the CO₂ adsorption is equivalent to 3.14% (0.71 mmol/g) out of the total 4.81% sorbed value, with the remaining 1.67% attributed to water adsorption. These results show that the presence of water has a positive effect, shifting the CO₂ sorption by 0.42% as compared to dry CO₂ sorption. It was observed that sorption patterns of dry CO₂, wet CO₂, and water on LDO look identical with the majority of the sorption taking place in the first few minutes of the experiment. Details of sorption values and percentage contributions of sorption taking place at different time intervals in these three cases are presented in Table 1. These results reveal in all of these cases that about 70% sorption occurs in the first 5 min, whereas a total of 75–85% sorption takes place within 10 min. On the other hand, sorption

Table 1. Weight Gained and Percentage Contributions of Dry CO₂, Wet CO₂, and Water Sorptions at Different Time Intervals

time (min)	Dry CO ₂		Wet CO ₂		Water	
	weight gained (%)	contribution (%)	weight gained (%)	contribution (%)	weight gained (%)	contribution (%)
2	1.80	66.2	2.52	52.3	0.73	43.7
5	2.13	78.3	3.25	67.6	1.26	75.4
10	2.31	84.9	3.65	75.9	1.41	84.4
20	2.47	90.8	4.15	86.3	1.50	89.8
30	2.56	94.1	4.39	91.2	1.55	92.8
40	2.63	97.7	4.55	94.5	1.61	96.4
50	2.69	98.9	4.69	97.5	1.66	99.4
60	2.72	100	4.81	100	1.67	100

kinetics slowed down for and adding only an extra 10% for the next 15 min. Hence, both CO₂ and water sorptions on LDO are quite rapid, and the process reaches nearly equilibrium in about 30 min. These fast sorption rates are particularly important in the context of CO₂ capture from flue gases using PSA or TSA processes.

Results presented in Figure 3 show the LDO sorption values of dry and wet CO₂ as a function of temperature. Dry CO₂ sorption follows a usual trend of a decrease from 200 to 400 °C but shows deviation with a lower sorption value at 100 °C as reported by Reddy et al.²² In the case of wet CO₂ sorption, a similar trend is also observed but with an increase in sorption values due to the presence of water. The sorption values show a significant jump in the presence of water at 100, 200, 300, and 400 °C. Wet CO₂ sorption at 100 °C shows an increase of 131% over dry CO₂, even though the net sorption values are lower than the values at 200 °C. Similar to dry CO₂ sorption, wet CO₂ also shows a maximum sorption capacity at 200 °C as already reported.²² Even though there is an overall decrease in dry and wet CO₂ sorption values at 300 and 400 °C, wet CO₂ sorption showed an increase of 96.9 and 210.7% over the dry values at these temperatures. This indicates that LDO takes up a considerable amount of water even at higher temperature in spite of low CO₂ sorption potential. Yong et al.¹⁷ and Ding et al.¹⁶ also investigated the influence of water on CO₂ sorption using LDHs. Both research groups reported weight gain of about 2.5% on the sorption at 300¹⁷ to 450 °C¹⁶. This is contrary to the present study where the difference between wet and dry CO₂ sorptions is much higher (Figure 3). The reason for this difference is that those groups used uncalcined LDH samples in their studies, which already contains an –OH network, whereas in the present study LDO is used. To differentiate CO₂ and water components from water-aided CO₂ sorption, an experiment was conducted by running wet CO₂ for about 60 min followed by dry CO₂. Figure 4 shows a 4.28% weight gain after running wet CO₂ for 50 min, whereas when switching to

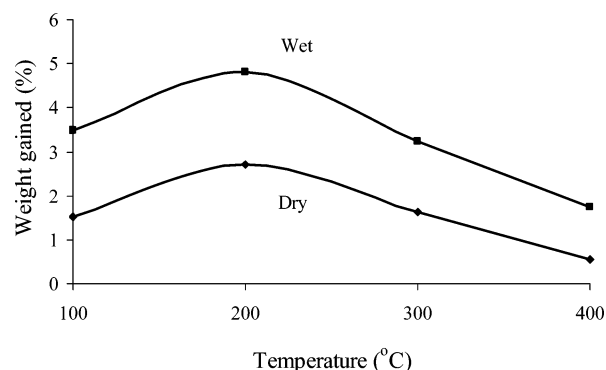


Figure 3. Comparison of wet and dry CO₂ sorption as a function of temperature.

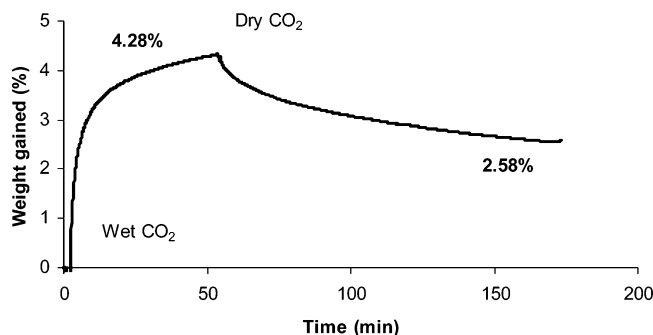


Figure 4. Sorption behavior of LDO on switching from wet CO₂ to dry CO₂.

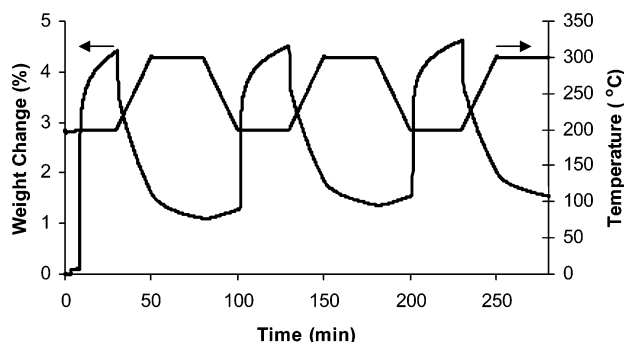


Figure 5. Wet CO₂ sorption and desorption behavior during temperature cycling.

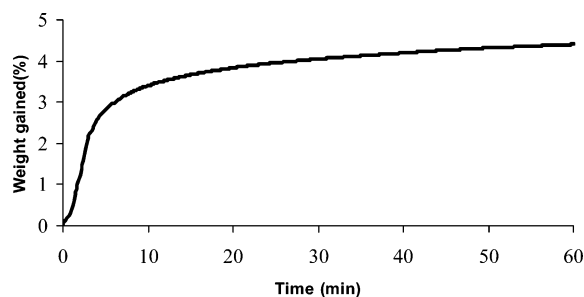


Figure 6. MG sorption on LDO in the presence of water at 200 °C.

dry CO₂ the weight started decreasing and reached 2.58% in about 2 h time, which is very close to the value observed in the dry CO₂ sorption case. The difference between these two values is 1.70%, which is again close to the water-sorption value of LDO. This suggests that additional sorption observed in the case of wet CO₂ is mostly due to water uptake.

CO₂ sorption and desorption behavior in the presence of water were studied using temperature cycling experiments. Results of wet CO₂ sorption and desorption of the LDO observed during three temperature cycles are presented in Figure 5. On average, the weight gain observed in each of the three sorption cycles is approximately 4.5%, whereas 75% of this value is desorbed after the first cycle. The desorption value, which shows a roughly 5–8% decrease at the initial cycles reached a constant value only after three cycles. This initial decrease in the desorption capacity during the first three cycles may be attributed to the stabilization of material and basic sites during the temperature swing. The overall consistency observed during these cycles reveals the recyclability and hydrothermal stability of the sorbent.

Sorption Using Mixed-Gas (14% CO₂) Feed. The influence of water present in the mixed-gas feed on CO₂ sorption was carried out on a wet-synthesis flue gas (i.e., mixed gas) containing 12% water. Figure 6 shows the sorption curve of LDO using wet mixed gas in a time testing span of 60 min.

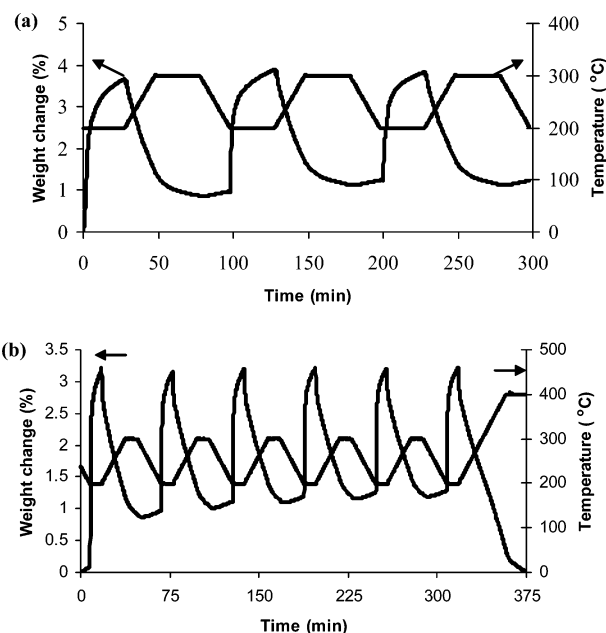


Figure 7. (a) MG sorption/desorption using temperature cycling in the presence of water and (b) sorption (200 °C) and desorption (300 °C) during six temperature cycles and regeneration.

These results follow almost identical sorption behavior as observed with the single gas above. Further, an additional weight gain of 1.41% occurred in the presence of water when compared to the dry sorption. The total sorption value equivalent to 0.90 mmol/g observed in this case under wet conditions is significantly high when compared to the earlier reported values (0.50 mmol/g).^{16,17} In addition, LDO's sorption potential remained constant even though CO₂ concentration is diluted by almost seven times in the mixed-gas testing. LDO did not show any proportionate decrease in the sorption capacity with the decrease in CO₂ concentration in the feed gas. Another important aspect of this study is the investigation of sorption/desorption potentials using temperature cycling in the presence of water. Part a of Figure 7 demonstrates the sorption and desorption capacity at 200 and 300 °C, respectively. Sorption that shows a weight gain of 3.6% was found to lose 75% of this weight upon desorption at 300 °C, indicating high reversibility of the system even in the presence of water. The sorption values remained constant throughout the experiment, whereas desorption values stabilized only after three cycles.

It is also observed that more than 70% of the total sorption takes place in the first 5 min and up to 85% sorption occurs within 10 min, similar to single-gas sorption trends. On the basis of this observation, cycle time was reduced to 10 min from 30 min and 6 sorption/desorption cycles were conducted followed by regeneration at 400 °C using wet mixed gas. The patterns obtained from this measurement are shown in part b of Figure 7. The sorption/desorption data extracted from this graph is listed in Table 2. These values indicate a weight gain of 3.22% during the first sorption cycle, which remained nearly constant for all of the six cycles. As expected, this sorption value is equivalent to 85% of the value observed in 30 min cycles shown in part a of Figure 7. These results demonstrated that LDO is quite efficient even with shorter time (10 min) cycles. Table 2 further reveals that evacuation of the sample during first cycle yielded up to 75% desorption, and this value slowly decreased to 67% after three cycles and remained almost constant after that showing the steady state. Finally, regeneration of the sample after six cycles resulted in almost 100% activation as seen from

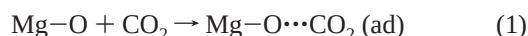
Table 2. Total and Reversible Sorption Values during 6 Cycles of 10 min Duration Using Wet Mixed Gas

cycle no.	total sorption (wt. %)	reversible sorption (wt %)	reversibility (%)
1	3.22	2.33	73
2	3.12	2.10	68
3	3.20	2.08	65
4	3.20	2.02	63
5	3.21	2.01	63
6	3.22	2.02	63

part b of Figure 7. Overall consistency and periodicity observed in the sorption/desorption curves during six cycles in the presence of water demonstrate the recyclability and hydrothermal stability of the LDOs even at low CO₂ partial pressure of the feed. Up to 100% reactivation achieved in this case proves the robustness of LDOs.

Structural Aspects of CO₂ Sorption and Decomposition.

Structure changes taking place during thermal decomposition of Mg–Al–CO₃ LDH are very important for understanding its CO₂ sorption properties. LDH completely loses interlayer water and dehydroxylates to a large extent upon heating to 400 °C, leading to mixed oxide formation with a 3D network. Although a fair amount of carbonate still remains in the structure, this gives a relatively high surface area and exposes sufficient basic sites on the surface.^{13,20} These basic sites favor reversible CO₂ sorption in a form as follows:



The interaction between the sorbed CO₂ and the basic sites seems to be mainly responsible for CO₂ sorption at 200 °C. Experimental results obtained in this study suggest that this interaction is not as weak like in the case of zeolite,²³ whereas it is not as strong as that in the case of alkali metal oxide.²⁴ These results strongly suggest the potential of the LDO application of CO₂ sorbent in flue-gas systems at temperatures around 200 °C and above. On the other hand, sorption of water by LDO at 200 °C suggests that it is more likely a chemisorption phenomenon rather than physical adsorption on the surface because it is more unlikely to happen at this temperature.^{15,16} LDO, which is an anhydrous mixed oxide solution of magnesium and aluminum, has a tendency to readily react with water. FTIR, XRD, and TGA traces recorded for the samples stored in atmosphere in our earlier studies have also clearly shown that LDO quickly takes up both water and CO₂.²² This also resulted in a partial reconstruction of the LDH structure as observed from the results presented therein. However, chances of holding any crystalline water at 200 °C are very low because it was clearly demonstrated in the thermal evolution studies of LDH that all of the physically bonded and crystalline water is removed at temperatures below 190 °C.^{19,22} Hence, it is very likely that these oxides react with water, forming hydroxides in the following manner.



In the presence of CO₂, they are expected to form respective bicarbonates, and this might be the reason for additional weight gain in the presence of water. It is observed from the sorption curves shown in parts a and b of Figure 2 that the difference between wet CO₂ and water sorption is almost equivalent to dry CO₂ sorption. This is also further evidenced from Figure 4, where switching from wet CO₂ to dry CO₂ led to a 1.70% decrease in the weight, which is nearly equal to the weight of

water sorbed by the LDO. This suggests that the weight loss observed in this case might be due to the decomposition of the bicarbonate, which is expected to dissociate in the absence of water and form carbonates in dry CO₂ as follows:



Conclusions

Presence of water in the feed did not influence the CO₂ sorption rate but had a positive impact on the CO₂ sorption efficiency of LDOs. It was found that the CO₂ sorption increased from 2.72% (0.61 mmol/g) to 3.14% (0.71 mmol/g) tested at dry and wet-gas conditions, respectively. Water is expected to react with LDO forming magnesium and aluminum hydroxides, which in the presence of CO₂ form bicarbonates without affecting the net CO₂ sorption. Regeneration of the compounds at 400 °C brought back more than 90% of the original sorption capacity. CO₂ sorption studies conducted using mixed gases (14% CO₂) have shown high sorption values even though the CO₂ concentration was diluted by almost seven times. LDO did not show any proportionate decrease in the sorption capacity with the decrease in CO₂ concentration in the feed gas. In the case of mixed gas, water did not hamper CO₂ sorption performance. LDO exhibited highly consistent sorption/desorption performance even after reducing the cycle time from 30 to 10 min. Temperature cycling in wet conditions demonstrated very high levels of desorption (75%), which reached an equilibrium value (67%) after initial stabilization. Shorter time cycles were found to be more effective in improving the overall efficiency of the process. These findings demonstrate the hydrothermal stability, recyclability, and regenerability of LDOs.

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